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Operando FTIR Investigation of Cathode Electrolyte Interphases Dynamic Reversible Evolution on $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$

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ABSTRACT: One of the keys to cycling stability of electrode materials is the formation of stable interface film on cathode materials, which called cathode electrolyte interphases (CEI). For Li/Mn-rich cathode especially, the high working voltage will cause extremely unstable electrolyte environment, becoming a challenge of stable interface film forming. In this work, an operando attenuated total reflection-fourier transform infrared (ATR-FTIR) technique is developed to real-time monitor the dynamic mechanism of CEI formation in carbonate-base electrolyte with or

without moderate additive tris (trimethylsilyl) borate (TMSB), which could promote the formation and stability of high quality CEI films when it charges to 4.8V. It is interesting that the components of CEI basically generate in the first cycle owing to EC priority decomposition. Besides, the presence of TMSB can suppress the decompose of ethylene carbonate (EC) in part, and modify the stability of CEI film. That is because TMSB containing electron-deficient boron atom can easily combine with electron-rich F^- and PF_6^- forming polyanion group initially, which will weaken electrostatic force between anionic groups and EC to reduce the concentration of EC on cathode surface and prevent EC continuously decompose at high voltage. XPS also verifies the presence of polyanion groups and they further participating in CEI forming. This work highlights the dynamical stability of CEI modified by moderate TMSB and its forming mechanism of this dynamical change during cycling characterized by operando ATR-FTIR, which paves a way for a better understanding of the complex and hard-characterized cathode interface reactions.

1. Introduction

As lithium ion batteries (LIBs) have aroused widespread concern as one of the most potential energy storage devices for their high specific energy, long cycle life, and good safety performance.¹⁻⁵ Deeper research for electrochemical mechanism of LIBs is indispensable. As we know, it will generate a mixed film of both inorganic and organic components along the surface of anode during electrochemical cycling in batteries,⁶ named solid electrolyte interphases (SEI) film.⁷ Same as SEI, researchers find there is a similar film generating on cathode surface during cycling, called cathode electrolyte interphases (CEI) film.⁸ The interface reactions and quality of SEI or CEI are related to electrochemical properties directly, so the exploration of the forming

mechanisms of SEI or CEI is quite meaningful.⁹ However, it is difficult to characterize the process of interface reactions through common measurements, because both SEI and CEI films are very sensitive to environment. So, to accurately explore interface reactions and interphase film formation, new characterization means, like *in situ* measurements should be developed.¹⁰⁻¹² Up to now, kinds of *in situ* characterization methods¹³ have been developed to study SEI film, including *in situ* TEM,¹⁴ AFM,¹⁵ XAS,¹⁶ Raman¹⁷ and so on.^{11, 18-20} Among these methods, *in situ* attenuated total reflection-fourier transform infrared (ATR-FTIR) spectroscopy measurement, on account of exciting molecular vibration, rotation, lattice mode or a combination of all these by total-reflection infrared band, is applied to analyze surface or interface groups, especially organic group in electrochemical system, which not only depend on active materials, but are related to electrolyte, like solvents, salts and additives.¹⁰ There are many works been done about SEI film inside various anode materials by *in situ* FTIR with different modes.²¹⁻²⁵ Nevertheless, the studies about CEI formation are far less than SEI film. What makes the interfacial research of cathode so difficult than anode owes to complex structures of cathode materials, unknown oxidized reactions of electrolyte under high voltage and hard-detected interactions between cathode and electrolyte. Recently, several works about cathode interface or CEI have been published. S. Jeong et al. put forward an additive lithium difluorobis(oxalato) phosphate ($\text{Li-PF}_2\text{C}_4\text{O}_8$) producing a fabulous CEI on LiCoO_2 at the first cycle to improve long cycling performance.²⁶ W. Zhao et al. enhancing both CEI and SEI on Ni-rich NMC76 and metal Li, respectively, by optimizing electrolyte (0.6 M LiTFSI + 0.4 M LiBOB + 0.05 M LiPF_6 dissolved in EC: EMC (4:6 by weight)) achieve significant increase on both cycling and rate performance.²⁷ And S. Jang et al. introduce an artificial CEI layer on Ni-rich NCM cathode by synthesizing lithium tetra(trimethylsilyl) borate (LTB) as a functional precursor. The insoluble LTB is introduced in slurry before coating on Al

foil, and forming a few nanoscale-based artificial CEI layers through the chemical reactions, which improves electrochemical properties obviously.²⁸ From this part, it can be known that CEI researches attract much attention in recent years. For decades, there have been several previous works studying cathode interface by *in situ* FTIR, including LiCoO₂,²⁹ LiFePO₄,³⁰ LiMn₂O₄ and LiNi_{1/2}Mn_{3/2}O₄³¹ from different pointcuts, such as solvent decomposing³² and choosing different lithium salts.³³ However, plenty of blanks still should be filled to give a better understanding of cathode materials, especially for new and potential materials, such as high-Ni ternary material, Li-rich material and so on. Among the new cathodes, Li/Mn-rich layered oxides delivering high specific capacity over 250 mAh g⁻¹ with high voltage (>4.5 V) and good thermal stabilities have attracted much attention as potential cathode of next generation.^{3, 34-36} But compared to common cathodes, the high work potential of Li/Mn-rich materials make their interface reaction more complicated owing to more different oxidation reactions generated, which causes unstable interface and ceaselessly capacity decaying. Therefore, it is necessary to figure out the mechanism of CEI forming and interface reaction for Li/Mn-rich materials to provide a theory support for further modification of both electrolyte and materials to improve the adaptability between them.

Li_{1.2}Ni_{0.2}Mn_{0.6}O₂, a promising Li/Mn-rich material free from noble metal,³⁷⁻³⁸ suffers voltage and capacity decay because of structural collapse started from particles surface during cycling, which forms unstable CEI. While stabilizing the surface of cathodes can be achieved by a simple method that is introducing expendable additives.³⁹ In this part, tris (trimethylsilyl) borate (TMSB), an additive with film-forming mechanism plays suitable properties for cathodes in high cut-off voltage.⁴⁰⁻⁴¹ To study how TMSB participating in CEI formation and further improving electrochemical properties in battery, operando ATR-FTIR is developed to characterize the environmentally sensitive CEI film and complex interface reactions.

In this work, electrolyte of 1M LiPF₆ in a mixture of ethylene carbonate (EC) / ethyl methyl carbonate (EMC) (v/v=1:1) with or without additive TMSB was studied for Li_{1.2}Ni_{0.2}Mn_{0.6}O₂ cathode material. The results of operando ATR-FTIR interestingly demonstrate that CEI basically generates during first cycle. The intensity of peaks from all components increases with the voltage in charging process, but displays irreversibility during discharging. However, in second cycle, it is found that the intensity of all peaks shows highly reversibility during both charge and discharge processes, which explains the CEI forming on cathode is dynamically stable in further cycles. Besides, operando ATR-FTIR results show that TMSB can combine with anionic groups and anions from lithium salt to reduce EC oxidized partially by cutting down electrostatic force between anionic groups and EC molecules. The components of CEI after cycling are analyzed by X-ray photoelectron spectroscopy (XPS), showing that CEI in optimized electrolyte gets solid because the adverse inorganic compounds such as Li₂CO₃, LiF reduced obviously. That means the obstacle during Li⁺ diffusing getting smaller, which is in accord with the increasing results of Li⁺ conductivity and diffusion rate. Through SEM and HRTEM, a more regular and thinner CEI can be observed on active particles when TMSB exists after 250 cycles. Finally, common electrochemical tests are performed to verify the positive effect of TMSB in battery optimizing and it is found long cycling performance at 1C is improved significantly. Thus, this work proves a simple way to optimize electrolyte and puts forward the reversible ability to give a fresh understanding for the sensitive CEI by operando ATR-FTIR.

2. Materials and Methods

2.1 Materials

Li_{1.2}Ni_{0.2}Mn_{0.6}O₂ was synthesis by coprecipitation method according to previous work.³⁷ The conventional electrolyte was a solution of 1 M LiPF₆ dissolving in dual-solvent of EC/EMC (v/v=1:2). Tris(trimethylsilyl)borate (TMSB) purchased from Shanghai Macklin Biochemical Co., Ltd. was added to conventional electrolyte by 0.25 %, 0.5 %, 1 %, 2 % mass ratio.

2.2 Materials characterizations

The morphology of Li_{1.2}Ni_{0.2}Mn_{0.6}O₂ before and after cycling was performed by scanning electron microscopy (SEM, JEOL JEM-700F) and high-resolution transmission electron microscopy (HRTEM, FEI Tecnai G2 F20). The samples prior to cycling were prepared by dispersing a little powder of pristine material in alcohol to get a uniform solution, and then dropping the sample on a micro grid using a pipette and drying after that. All the cycled Li_{1.2}Ni_{0.2}Mn_{0.6}O₂ was extracted from 2032 coin-type cells after 250 cycles. The electrodes disassembled from the cells were washed by EMC and alcohol three times, respectively. Then the material on electrodes was scraped and ground with N-methyl-2-pyrrolidinone (NMP) by mortar. The mixture was filtered and washed by alcohol three times, and then dispersed into alcohol uniformly. After that, the micro grid with cycled sample was prepared by the same way as uncycled powder. X-ray photoelectron spectroscopy (XPS) data was obtained on a Perkin-Elmer PHI 5000C ESCA system equipped with a twin anode Mg K α (1253.6 eV) X-ray source. Electrodes after one active cycle at 0.1C and 10 cycles at 1C in both electrolyte with and without TMSB were obtained to test CEI film properties by XPS.

2.3 Electrochemical measurements

Active material, black carbon (super p) and binder (Polyvinylidene fluoride, PVDF) were mixed by mass weight ratio 84: 6: 10 in N-Methyl pyrrolidone (NMP). The mixed slurry was then coated

on an aluminium foil with certain thickness 15 μ m, which was dried, rolled and pouched to electrodes with 15 mm diameter after coated. The loading mass of active material was controlled about 1.5 mg cm⁻² by the amount of solvent NMP and coating thickness. The electrodes were finally moved into Argon filled glove box (H₂O < 0.5 ppm, O₂ < 0.5 ppm) after weighed and dried at 120 °C overnight. Electrochemical measurements were tested using 2032 coin-type cells. The cycling life and rate capability were performed by galvanostatic charged and discharged progress ranging from 2.0 to 4.8 V with LANHE CT-2001A at room temperature. The calculation of current density: 1C = 250 mAh g⁻¹. Electrochemical impedance spectra (EIS) was evaluated by a CHI 660E electrochemical workstation and PARSTAT Multichannel (type PMC-1000 form Princeton Applied Research) in the frequency range from 10 kHz to 10 MHz. Cyclic voltammetry curve was measured by a CHI 660E electrochemical workstation in the frequency range from 100 kHz to 10 MHz. The range of test voltage was 2.0-4.8 V and the sweep speed was 0.5 mV/s.

2.4 Operando ATR-FTIR measurements

Operando ATR-FTIR experiments were performed with a three-electrode electrochemical cell made of Teflon. Working electrode was metal Pt with active material slurry coated on it, and both of counter and reference electrodes were metal Li. The working electrode was dried at 120°C in vacuum oven overnight, then pressed against a 45°-cut CaF₂ prism. The electrode could be adjusted by camshaft to form a thin layer electrolyte between prism and active material. Two mirrors were used to reflect infrared into prism, and then bring the signals into detector, also light path and infrared strength could be adjusted by mirrors (Figure S4). To detect the potential-dependence oxidation reaction at the first cycle, a CV test for three-electrode *in situ* cell was performed from 2.0 V to 4.8 V, scan speed 0.5 mV/s. FTIR spectra collected every 0.1 V during CV testing, namely

200 s per spectrum automatically. After the first cycle, the cell was standing until voltage getting stable. Then the second cycle was tested under the same test condition to check if there were new species generating after the first cycle. All of the operando ATR-FTIR was measured by a ThermoFisher Nicolet 6700 equipped with a mercury cadmium-telluride (MCT) detector. All spectra were obtained with a resolution of 4 cm^{-1} and every spectrum was acquired by adding a total of 64 scans.

3. Results and Discussion

3.1 TMSB enhance electrochemical properties of $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$

Both electrolyte conductivity and electrochemical stable window are key parameters for judging excellent electrolyte in a lithium ion battery, so we first evaluate the effect of TMSB for these two performances (Figure S1, Figure 1A-1C) and optimize TMSB content (Table S1 and Figure S2). According to the results, it is found that electrolyte with 0.5% TMSB show a smaller intercept (1.69 ohm) compared to conventional electrolyte (2.44 ohm) as shown inset in Figure 1B, which means a better conductivity after TMSB added. Moreover, the electrolyte with 0.5% TMSB possesses the highest conductivity of $5.41 \times 10^{-4}\text{ S/cm}$ calculated by electrochemical impedance spectra (EIS) of electrolytes (Table S1), much higher than that of conventional electrolyte ($3.8 \times 10^{-4}\text{ S/cm}$), seen in Figure 1C. Besides, its wider electrochemical stability window of 5.27 V indicates that the additive can improve the stability of electrolyte to some extent, as shown in Figure 1A. Hence, for a more convenient study, 0.5% TMSB-containing and conventional (0% TMSB) electrolyte are selected as comparisons in this work. $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$ is synthesized successfully (Figure S3) and its EIS in two electrolytes are shown in Figure 1D. It is obvious that impedance of $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$ reduces from $94\ \Omega$ to $62\ \Omega$ after TMSB added. The corresponding diffusion

coefficients of Li^+ significantly increases from $2.65 \times 10^{-14} \text{ cm}^2 \text{ s}^{-1}$ to $4.94 \times 10^{-14} \text{ cm}^2 \text{ s}^{-1}$ (Table S2) based on calculation of electrochemical impedance. The Figure 1E shows the rate capability from low current density 0.1C to high 5C then back to 0.1C. As the current density increasing, the difference between these two electrolytes gets larger. At 2C, 0.5% TMSB-containing electrolyte can still deliver discharge specific capacity of 141.0 mAh g^{-1} , which is far higher than that of conventional electrolyte 93.5 mAh g^{-1} . Even at high current density 5C, the capacity of conventional electrolyte is only 2.4 mAh g^{-1} , while as for 0.5% TMSB-containing electrolyte, capacity about 49 mAh g^{-1} is still retained. In addition, after the current density returns to 0.1C, the capacity retention of TMSB-containing electrolyte is 94.5% (compared to the discharge capacity at initial 0.1C during the rate capability test), which is higher than that of conventional electrolyte, 88.9%.

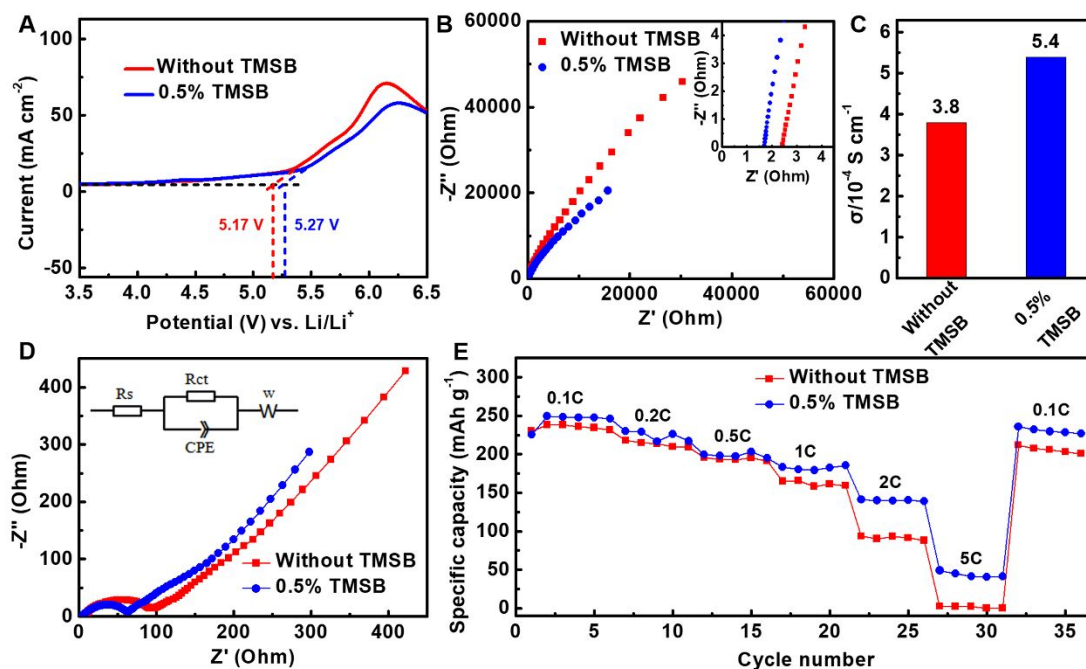


Figure 1. Electrochemical properties of two electrolytes and half cells with $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$ in two electrolytes. (A) The electrochemical window of electrolytes. (B) The conductivity of electrolytes. (C) Conductivity values of two electrolytes calculated by (B). (D) Electrochemical impedance spectra (EIS) before cycling, and the corresponding equivalent circuit diagram was shown inset. (E) Rate capability from 2.0 V to 4.8 V (vs. Li/Li^+) at different current density from 0.1C to 5C, and then back to 0.1C.

In general, TMSB can increase electrochemical stability window and electrolyte conductivity of electrolyte, and reduce impedance of battery obviously. That may result from the formation of polyanion group $(\text{TMSB-PF}_6)^-$, because TMSB contains electron-deficient boron atoms, which tends to attract electron-rich group PF_6^- , leading to more Li^+ in LiPF_6 released. So, the interface

compatibility between electrolyte and electrode material is improved, resulting in interfacial impedance reducing significantly, and rate capability increasing.

3.2 Operando ATR-FTIR real-time monitoring CEI formation and evolution

To explore the dynamic mechanism of the additive TMSB participating in CEI forming, operando ATR-FTIR is developed to study nonequilibrium states of CEI film (schematic of *in situ* cell can be found in Figure S4). The signals of ATR-FTIR are collected during the CV test within first two cycles. Figure 2 shows the operando ATR-FTIR spectra in the corresponding charge progress during the CV tests. There are less than 10 cm^{-1} difference in all these peaks between common ATR-FTIR (Figure S5) and operando ATR-FTIR spectra, which may owe to different test environments. Figure 2A and 2D are charging process of the first cycle in conventional and TMSB-containing electrolyte, respectively, and the corresponding CV curves are shown in Figure 2C and 2F. The magnifications of Figure 2A and 2D ranging from 1500 to 1700 cm^{-1} are shown in Figure 2B and 2E, respectively. It can be seen that the intensity of all peaks gets stronger with the voltage increasing from open circuit potential (OCP) to 4.8 V. Compared to conventional electrolyte, the peak at 1646.9 cm^{-1} corresponding C=C group in vinyl ester disappears after 0.5% TMSB added. Since vinyl ester is a decomposed product of EC, this illustrates TMSB can suppress the decomposition of EC.

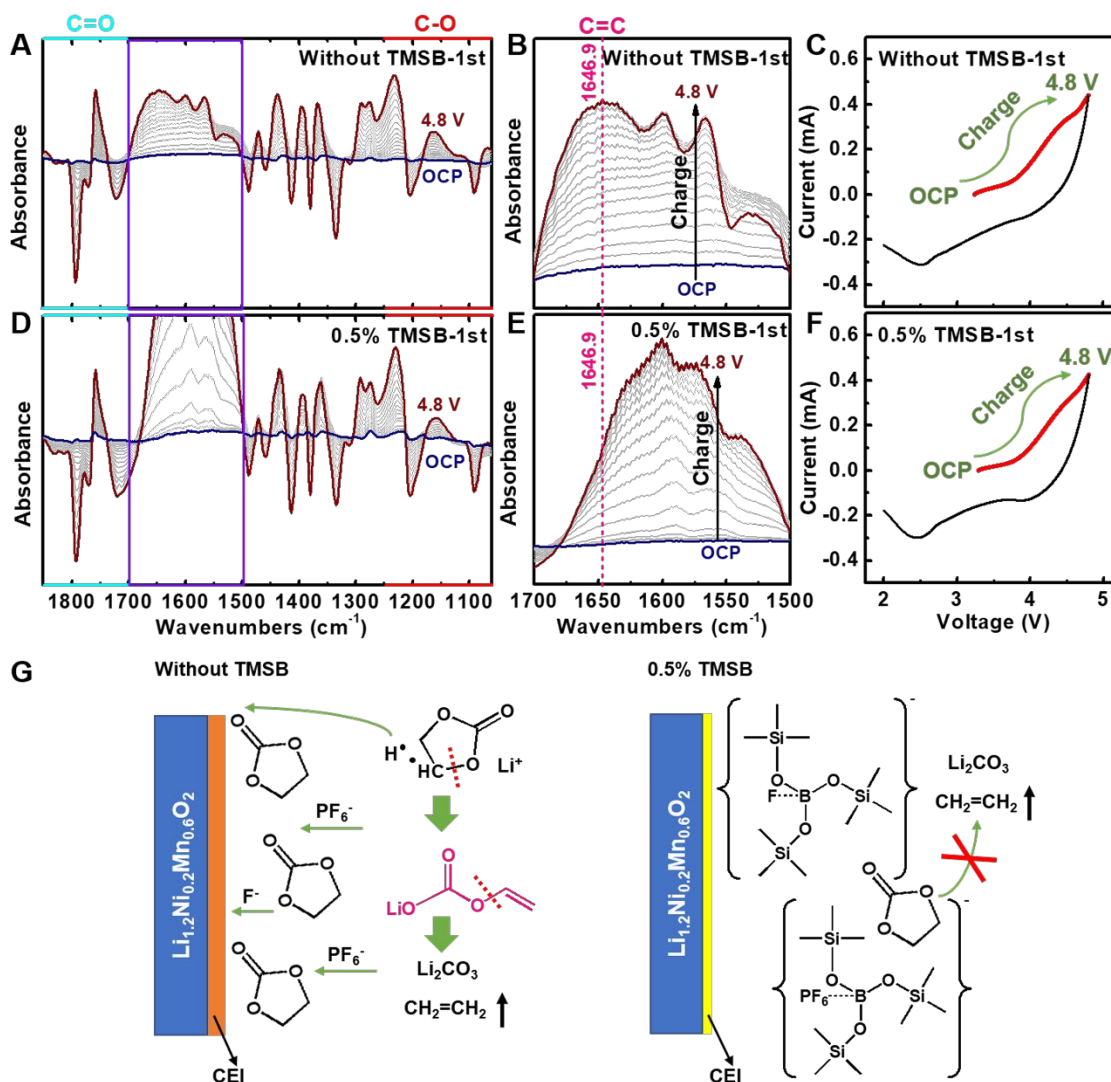


Figure 2. Operando ATR-FTIR spectra and schematic of relevant reactions at first charging process. (A) Operando ATR-FTIR spectra of first charging process in conventional electrolyte. (B) The magnification of purple region in (A) from 1500 to 1700 cm^{-1} . (C) Corresponding CV curves in conventional electrolyte. (D) Operando ATR-FTIR spectra of first charging process in 0.5% TMSB-containing electrolyte. (E) The magnification of purple region in (D) from 1500 to 1700 cm^{-1} . (F) Corresponding CV curves in 0.5% TMSB-containing electrolyte. (G) Schematic of relevant interface reactions at active material surface in conventional and 0.5% TMSB-containing electrolyte, respectively.

In electrolyte system, although the structure of EC is more stable than linear carbonates, such as DEC, DMC, EMC and so on, EC will decompose before these linear carbonates in a general way. Resulting from the larger polarity, EC molecule has strong electrostatic force with the anionic groups in lithium salt, which tends to migrate to cathode surface with anions during charge easily.⁴² So, EC is always oxidized before linear carbonates. At high voltage, the dehydrogenation reaction will occur in EC molecule when electrons lost, then the unstable EC molecule is easy to decompose forming unsaturated compound, which is an intermediate and will generate ethylene and Li_2CO_3 further. However, when TMSB is added in the system, electron-deficient B atom can combine with anions and anionic groups forming a larger anionic group mentioned earlier. Then electrostatic force between EC and anionic groups gets weaker because of the larger molecule and lower electronegativity of anionic group, so that the interaction between EC molecule and anions of lithium salt gets weaker as well. That means less EC molecule will be brought to cathode surface, so the decomposition of EC can be suppressed to some extent. All the structure of these molecules mentioned above is shown in Figure S6. And the schematic explaining the whole reaction process in conventional and 0.5% TMSB-containing electrolyte is shown in Figure 2G.

Figure 3A and 3D are the operando ATR-FTIR spectra of first discharge process from 4.8 V to 2.0 V (Figure 3C and 3F) in the same region as Figure 2A and 2D in conventional and TMSB-containing electrolyte, respectively. It is seen that the location of peaks in two electrolytes are almost the same. But the intensity of most peaks appearing in charging process keeps rising in conventional electrolyte during discharge process. On the contrary, in TMSB-containing electrolyte, the intensity rises a little or remain unchanged, especially in the purple region, shown in Figure 3B and 3E. That means the CEI in TMSB-containing electrolyte is more stable. It forms during charging process, and maintain the species and concentrations when the test goes to

discharge. Unlike TMSB-containing electrolyte, intensity continually rising means the concentrations of components in conventional electrolyte keep changing, namely reactions sustain generating, which may result in a thicker CEI forming on active material. In conclusion, the role of TMSB is not only reducing EC oxidized, which is the main factor leading to an unstable environment for electrolyte, but also making CEI film stable in first cycle to modify the quality of CEI.

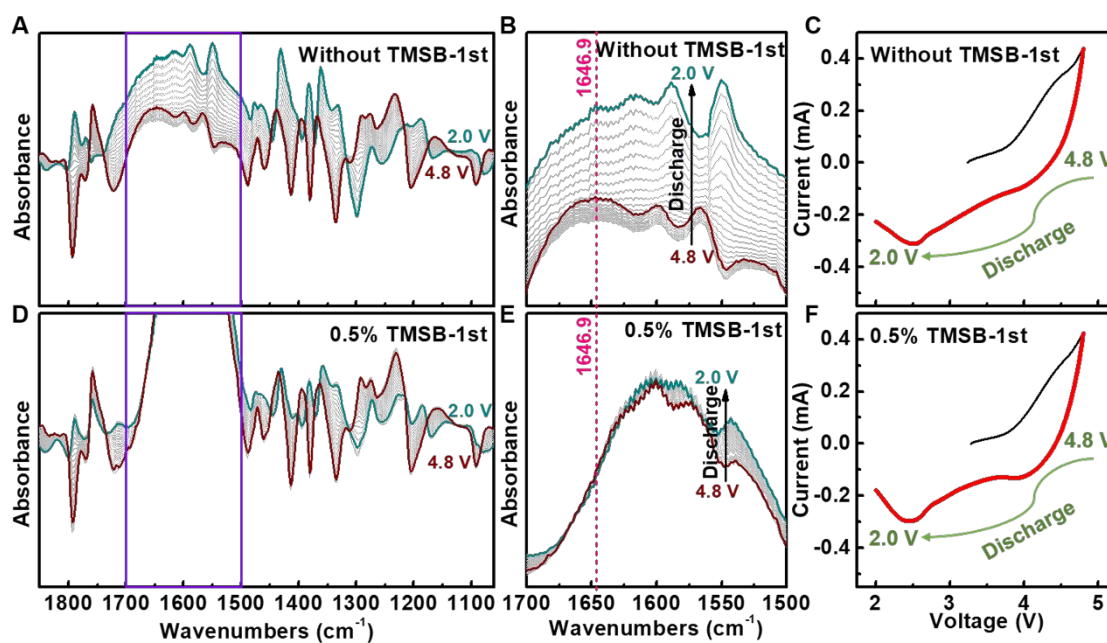


Figure 3. Operando ATR-FTIR spectra at first discharging process. (A) Operando ATR-FTIR spectra of first discharging process in conventional electrolyte. (B) The magnification of purple region in (A) from 1500 to 1700 cm^{-1} . (C) Corresponding CV curves in conventional electrolyte. (D) Operando ATR-FTIR spectra of first discharging process in 0.5% TMSB-containing electrolyte. (E) The magnification of purple region in (D) from 1500 to 1700 cm^{-1} . (F) Corresponding CV curves in 0.5% TMSB-containing electrolyte.

After the first cycle, *in situ* cell is standing until the voltage gets stable again. Then the background signal is collected again before the second cycle. The purpose is to check the stability of CEI forming in first cycle during further cycles. From operando ATR-FTIR results in second cycle (Figure S7 and S8), peaks appear at same wavenumbers as in first cycle, and all the components generating in two electrolytes display highly reversible during charging and discharging with no difference. That means CEI is ion conductive, and the concentrations of components change with ion exchanging during cycling. The component and morphology of CEI after cycling will be discussed next by common characterization to verify the *in situ* results.

3.3 The stable mechanism of CEI

XPS and HRTEM are employed to investigate the components and morphology of CEI after cycling as seen in Figure 4. Figure 4A, 4C and 4E are the XPS spectra of pristine $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$, cycled electrode in conventional and 0.5% TMSB-containing electrolytes, respectively, after 11 cycles (containing 1 cycle at 0.1C and 10 cycles at 1C). While Figure 4B, 4D and 4F are the corresponding HRTEM images after 250 cycling.

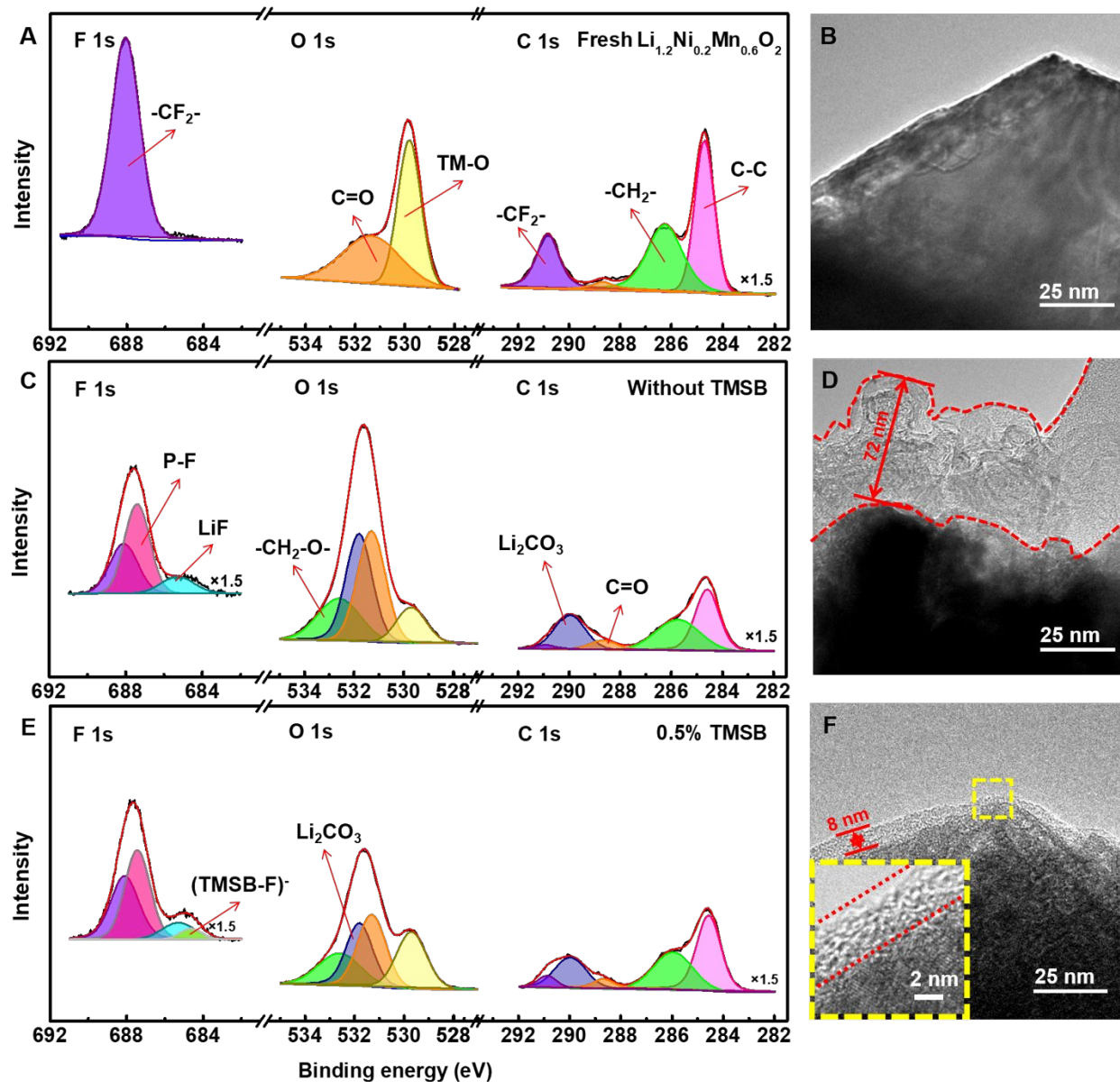


Figure 4. X-ray photoelectron spectroscopy (XPS) spectra and high-resolution transmission electron microscopy (HRTEM) images. (A, C and E) XPS spectra of C 1s, O 1s and F 1s for (A) fresh $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$ electrodes and cycled electrodes in (C) conventional and (E) 0.5% TMSB-containing electrolyte before and after 11 cycles (1 cycle at 0.1C for active and 10 cycles at 1C). (B, D and F) HRTEM images of (B) pristine and cycled $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$ after 250 cycles in (D) conventional and (F) 0.5% TMSB-containing electrolyte.

As seen in C 1s spectra of three samples, peaks at 284.6 eV, 286.0 eV and 291.1 eV are C-C (from carbon black), -CH₂- and -CF₂- group (both from PVDF), respectively. The intensity of all the peaks in pristine material (Figure 4A) are higher than the cycled ones (Figure 4C and 4E). This results from the formation of CEI with certain thickness during cycling, which will reduce the detected ability of bulk components. The clear edge and sharp of fresh particles can be observed in HRTEM (Figure 4B) and SEM images (Figure S9A and D). After cycling, new peaks can be observed at 290.0 eV and 288.7 eV, which belongs to Li₂CO₃ and C=O, and both come from the decomposition of EC. After active material cycled in the conventional electrolyte, there is a very strong peak of Li₂CO₃ observed in Figure 4C, while the signal of -CF₂- from PVDF significantly reduces because of a thick CEI forming. Moreover, from Figure S9B and E, it can be seen the shape of particles became unclear and they are gathering together obviously, and there is an irregular CEI about 72 nm thick covering the particle surface shown in HRTEM (Figure 4D). On the contrary, with 0.5% TMSB added, the -CF₂- from PVDF can be detected clearly seen in Figure 4E, meaning a thinner CEI, which is certified in Figure 4F. A thinner CEI film about 8 nm forms along with the particle edge. From the boost of yellow region in Figure 4F, it shows that the thinnest CEI is less than 5 nm and the clear crystal lattice can be observed in the bulk. SEM image shows the clear edge of particles with the unchanged size compared to uncycled ones (Figure S9C and F). Furthermore, the intensity of Li₂CO₃ peak decreases markedly and the peak of C=O is also slightly smaller. All these results confirm that the addition of TMSB can suppress the decomposition of EC and make the CEI solid which is in according with operando ATR-FTIR results.

In O 1s curves of fresh Li_{1.2}Ni_{0.2}Mn_{0.6}O₂ electrode seen in Figure 4A, C=O from residual solvent NMP (~531.3 eV) and TM-O (~529.7 eV, TM=transition metal) peaks can be observed. After

cycling, the peaks of $-\text{CH}_2\text{-O}-$ and Li_2CO_3 emerge at 532.6 eV and 531.8 eV (Figure 4C and 4E), respectively. The reason why intensity of C=O peak getting stronger is that partial EC has decomposed. Moreover, the obvious decreasing of TM-O peak is contributed to the thick CEI forming in conventional electrolyte shown in HRTEM images. However, the decomposed product significantly reduces with TMSB added (Figure 4E), which demonstrates the existence of additive TMSB will restrain EC decomposing, consistent with C 1s spectra. Besides, the slight increase of TM-O peak illustrates a thinner CEI formed in TMSB-containing electrolyte. From F 1s spectra of pristine and cycled $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$ electrodes, the peak at ~ 688.1 eV is $-\text{CF}_2-$ from PVDF, and it gets obviously smaller after cycling in both conventional and TMSB-containing electrolyte because of the formation of CEI. Furthermore, there are new peaks appearing at ~ 687.4 eV and ~ 685.3 eV after cycling, which are identified as P-F and LiF, respectively. These two inorganic components come from the decomposing of LiPF_6 , which leads to the unstable electrolyte environment.⁴³ Especially LiF, considered as an insoluble product, will bring conductive Li^+ into CEI, forming a part of “dead” Li^+ , resulting in less free Li^+ and larger charge transfer resistance. Different from conventional electrolyte, 0.5% TMSB-containing electrolyte displays a smaller peak of LiF in Figure 4E. Interestingly, there is a new little peak appearing at lower binding energy ~ 684.6 eV, far lower than that of F 1s. That means the peak comes from an electron-rich group, making the binding energy smaller than unbind F atom. So, the electron-rich group should be $(\text{TMSB-F})^-$ as mentioned before, which means TMSB can participate in CEI forming by two ways: one is chemical reactions by B-O bond breaking; the other is electron-rich group entering the CEI directly. The same regulation can be demonstrated in Figure S10, corresponding to B 1s spectra. The palpable test noise results from the low content of this additive. The B-O peak at ~ 191 eV, comes from TMSB molecule. The small peak at higher binding energy ~ 194.3 eV is considered as

the signal of (TMSB-F)⁻ because of the high electronegativity of F⁻. Both F 1s and B 1s spectra of TMSB-containing electrolyte testify the existence of the electron-rich group (TMSB-F)⁻, which could reduce the formation of LiF and release the “dead” Li in CEI. Figure S11 is P 2p spectra of cycled Li_{1.2}Ni_{0.2}Mn_{0.6}O₂ with and without TMSB, respectively. The weak signal from two main peaks are Li_xPF_y and Li_xPO_yF_z at ~136.4 eV and ~133.6 eV, respectively. Both are the reaction products of LiPF₆.⁴³ After TMSB added, the peak of Li_xPF_y reduces slightly, explaining that the harmful side reaction is suppressed at a certain extent. Besides, from the F 1s spectra, the intensity of P-F peak gets weaker after TMSB added, explaining the P-containing components decreasing markedly.

To intuitively observe the change after TMSB added, digital photos of cycled electrodes and metal lithium are taken shown in Figure S12. After 200 cycles, both electrode and lithium are cleaner in 0.5% TMSB-containing electrolyte than in conventional electrolyte, which explains the migration of transition metal ions are alleviated. That means the CEI modified by TMSB is protective for active materials, which is also verified by the maintained crystal structure and morphology of Li_{1.2}Ni_{0.2}Mn_{0.6}O₂ particles as mentioned in HRTEM and SEM images. The thin but stable CEI can keep active material from the corrosion of electrolyte and reduce the solution of transition metal ions at some extent and further effectively preventing the bulk structure collapse, which will be beneficial to a better cycle performance of Li_{1.2}Ni_{0.2}Mn_{0.6}O₂.

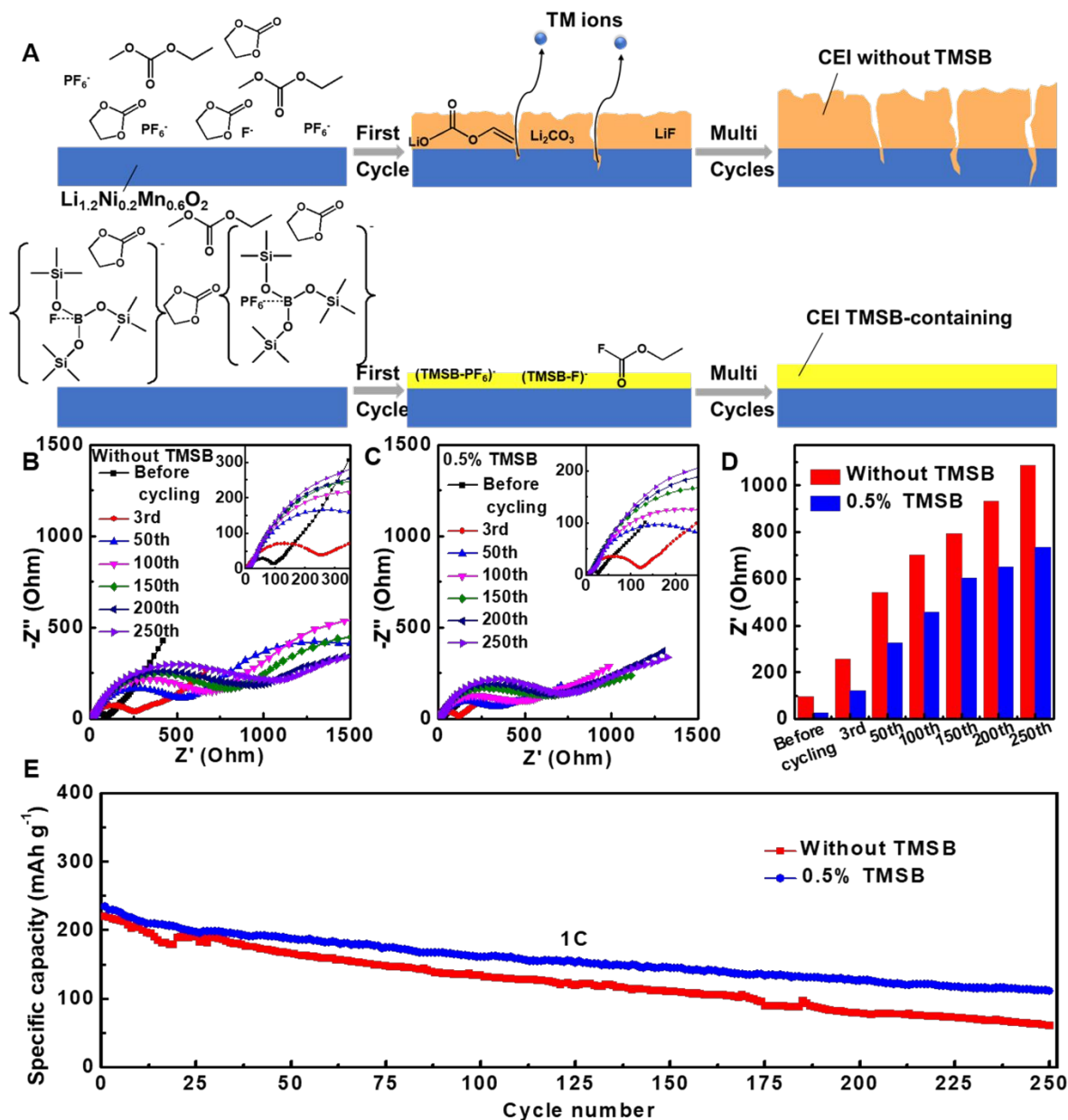


Figure 5. Schematic of CEI formation and electrochemical properties of $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$. (A) Schematic of CEI formation during cycling in conventional and TMSB-containing electrolyte. (B-D) Electrochemical impedance spectra (EIS) after 3rd, 50th, 100th, 150th, 200th, 250th cycles at 1C in (B) conventional electrolyte and (C) 0.5% TMSB-containing electrolyte. (D) Bar graph of impedance value in two electrolytes. (E) Long cycle performance from 2.0 V to 4.8 V (vs. Li/Li^+) at high current density of 1C.

Figure 5A shows the whole mechanism of CEI formation with and without additive TMSB. Generally, larger polarity of EC molecules makes them easily diffusing to cathode surface with anions by strong electrostatic force during charge process. So, cathode surface is EC/anions-rich before CEI forming, which makes EC easily decomposed. However, with TMSB added, the affect will impair due to the generation of polyanion group (TMSB-PF₆)⁻ and (TMSB-F)⁻ with weaker electrostatic force and less electronegative. As a result, EC decomposing is suppressed effectively. Moreover, TMSB can participate in CEI formation by electrochemical reactions or directly enter CEI in the form of polyanion group to optimize the quality of CEI. Its higher HOMO level (-6.749 eV)⁴⁴ than carbonate solvent (-11.905 eV for EC⁴⁵ and -11.541 eV for EMC) means lower oxidation potential,⁴⁶ which offers opportunity to participate in the formation of high-class CEI film before solvent decomposing to protect active material from erosion of electrolyte, and alleviate the capacity and voltage fading of Li_{1.2}Ni_{0.2}Mn_{0.6}O₂.

To further verify the effect of TMSB to electrochemical stability, electrochemical impedance spectra (EIS) after different cycles is measured. When cells discharging to 2.0 V after 3rd, 50th, 100th, 150th, 200th, 250th cycle, EIS is got to evaluate the conductivity and interfacial stability of the whole system, as shown in Figure 5B and 5C. EIS is consist of two parts: one is the semicircle at high frequency region, whose diameter represents the resistance of Li⁺ through interface film, which is CEI in this system; the other is the slope at low frequency region, which is the resistance of Li⁺ diffusing in the bulk active materials. Compared Figure 5B with 5C, a bar graph Figure 5D is got. From this graph, the impedance value of conventional electrolyte increases from 100 Ω to 250 Ω after only 3 cycles. While in TMSB-containing electrolyte, the impedance value increases more slightly from 30 Ω to 120 Ω. Even after 250 cycles, the impedance value of TMSB-containing electrolyte reaches to 550 Ω, which is far less than that of conventional electrolyte, over 1000 Ω.

These results demonstrate that the conductivity of electrolyte is improved with the existence of TMSB, identified by electrochemical and conductivity properties discussed earlier. Furthermore, the interface of cathode is more stable after TMSB added, contributed to a solid CEI on active particles. That means the CEI formed with TMSB is stronger and with higher quality. Finally, although first charge/discharge curves and CV curves show little difference before and after TMSB added (Figure S13 and S14), the capacity retention of the TMSB-containing electrolyte after 250 cycles is 47.7%, 18.4% higher than conventional electrolyte (29.2%) which can be got from long cycling performance at 1C (Figure 5E). The result demonstrates the role of TMSB to modify the CEI and further stabilize the interface between electrolyte and active material. In a word, the decomposing and other effects of TMSB at high voltage not only can suppress EC decomposing, but is beneficial to form an ion-conducted film to protect active material, so that a better electrochemical property can be achieved.

4. Conclusions

In conclusion, an operando ATR-FTIR was developed to study interface reactions between Li/Mn-rich cathode material $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$ and carbonate-base electrolyte before and after film-forming additive TMSB added, as well as propose the mechanisms of CEI formation and modification by TMSB. Experimental results showed TMSB was inclined to combine with electron-rich ions, such as F^- and PF_6^- , so that EC decomposing owing to strong electrostatic force with anion and anionic group was suppressed and more “dead” Li of LiF locked in CEI could be released in solution. That resulted in higher conductivity, faster Li^+ diffusion rate and lower Li^+ transmission resistance, so the rate capability in 0.5% TMSB-containing electrolyte increased significantly. In addition, operando ATR-FTIR results also shown TMSB is beneficial to a more stable CEI forming, and HRTEM and XPS results demonstrated that a thinner and more stable CEI

would form after TMSB added because TMSB would participate in the formation of CEI. The stable interface between active material and electrolyte not only could reduce resistance during cycling, but protect cathode from erosion of HF and relieve Mn ion migrate to anode, which would decrease structural collapse, so that better long cycling performance could be achieved. Herein, operando ATR-FTIR is developed to verify the effect mechanism of TMSB, an additive introduced to optimize electrolyte for $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$ cathode and make the material maximize its performance. This *in situ* measurement create an opportunity to further explore the interface reactions during cathode cycling. That would provide theoretical supports for $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$ modification and its electrolyte optimizing in next step. This work put forward a different understanding about complicated cathode surface, filling the research gap of CEI and interface reactions of new cathode materials and giving a deeper understanding of materials for their further modification. And operando ATR-FTIR tests can be used only or combined with other *in situ* measurements, like *in situ* Raman to study interface actions on other potential cathodes and give a systematic theoretical support about CEI.

Supporting Information.

Schematic of electrolyte electrochemical tests; cycling performance at low current density of 0.1C in electrolytes without and with 0.25%, 0.5%, 1% and 2% TMSB; schematic diagram of the *in situ* pool; ATR-FTIR spectra of solvent and several electrolytes with different components; molecular formula of solvents, decomposing product and additive; operando ATR-FTIR spectra of second cycle at range of 1850-1060 cm^{-1} ; operando ATR-FTIR spectra of second cycle at range of 3000-1850 cm^{-1} ; SEM images of pristine and 250 cycled $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$; XPS spectra of B 1s for $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$ electrode after 11 cycles in 0.5% TMSB-containing electrolyte; XPS spectra of P 2p for $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$ electrode after 11 cycles in conventional and 0.5% TMSB-containing

electrolyte; digital photos of cycled electrodes; the charge/discharge curves of first cycle of $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$ in conventional and 0.5% TMSB-containing electrolyte from 2.0V to 4.8V (vs. Li/Li^+); the cyclic voltammogram curves of first cycle of $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$ in conventional and 0.5% TMSB-containing electrolyte from 2.0V to 4.8V (vs. Li/Li^+); the conductivity of electrolytes with different TMSB content ; kinetic parameters of $\text{Li}_{1.2}\text{Ni}_{0.2}\text{Mn}_{0.6}\text{O}_2$ in electrolyte with and without TMSB. (PDF)

Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

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